

CSC 212: Data Structures and Abstractions

02: Introduction to Analysis of Algorithms (part 1)

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Analysis of algorithms

Problems, algorithms, programs

- Problem
 - ✓ a computational problem is a formalized task requiring a solution
 - ✓ well-defined input specification, expected output, formal constraints
- Algorithm
 - ✓ a precise, unambiguous sequence of computational steps
 - ✓ essential properties:
 - **correctness**: produces valid output for all valid inputs
 - **finiteness**: terminates after a finite number of steps
 - **determinism**: produces consistent results
 - **feasibility**: each step is executable
- Program implementation
 - ✓ concrete realization of an algorithm using a programming language

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Q2-1

- A recipe step says: “Randomly decide whether to add salt.” Which property of a well-defined algorithm does this violate?
 - A. Correctness
 - B. Finiteness
 - C. Determinism
 - D. Feasibility
 - E. None of the above

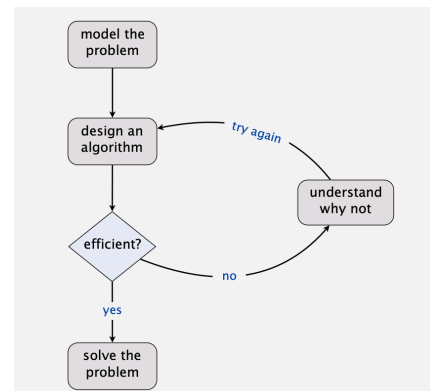
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Answer

- C) Determinism
 - ✓ an algorithm must produce the same output every time it runs on the same input
 - ✓ a coin-flip step makes the result non-reproducible
 - not correctness, it might still produce a valid dish
 - not finiteness, it terminates
 - not feasibility, flipping a coin is perfectly executable

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Developing a usable algorithm



[COS 226 lectures, Princeton University]

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Analysis of algorithms

- Definition
 - ✓ the branch of CS that studies the **efficiency of algorithms** in terms of **time** and **space** complexity
- Objectives
 - ✓ **performance prediction**: estimate running time growth with input size
 - ✓ **resource optimization**: identify bottlenecks; minimize CPU time and memory
 - ✓ **comparative evaluation**: provide standardized metrics to compare algorithms that solve the same problem

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Why it matters?

- Scientific classification of algorithms
 - ✓ groups algorithms by computational characteristics; complexity classes guide design and approximation strategies
- Performance prediction and resource utilization
 - ✓ anticipates behavior before implementation, avoiding costly rewrites
- Time and space complexity
 - ✓ critical where processing power or memory is constrained
- Theoretical guarantees
 - ✓ provides upper-bounds for reliable, predictable system design

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Two approaches

- Empirical analysis (today)
 - ✓ implement the program
 - ✓ systematic testing across varied input sizes
 - ✓ statistical analysis of the measurements => **hypothesis**
 - ✓ validation through **prediction**
- Theoretical analysis (next lecture)
 - ✓ **mathematical modeling** of time and space complexity
 - ✓ no implementation, no execution

These are not rivals. They are the two halves of the scientific method applied to code.

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Q2-2

- You want to compare two sorting algorithms without writing or running any code. Which approach are you using?
 - A. Empirical analysis
 - B. Theoretical analysis
 - C. Both, simultaneously
 - D. Neither, this isn't possible
 - E. None of the above

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Answer

- B) Theoretical analysis
 - ✓ theoretical analysis builds a mathematical model of cost (counting operations as a function of n) and needs no implementation or execution
 - ✓ empirical analysis, by contrast, requires an actual program you run and time

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Empirical analysis

Example: evaluating a polynomial

- A polynomial of degree n is written:

$$p(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n = \sum_{i=0}^n a_i x^i$$

- The task:** given given the coefficient array $a[0..n]$ and a value x , compute $p(x)$

Two programs follow. Both take the same array of coefficients and the same x . Both return the same number. One of them does dramatically more work than the other, can you tell which by glancing at them?

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Solution 1 (rebuild each power)

```
double poly1(const double a[], int n, double x) {
    double p = 0.0;
    for (int i = 0 ; i <= n ; i++) {
        double xp = 1.0;
        for (int j = 1 ; j <= i ; j++) {
            xp *= x;
        }
        p += a[i] * xp;
    }
    return p;
}
```

Check the redundant computation with $n=4$. The inner loop rebuilds x^i from scratch every time, having just computed x^{i-1} one iteration earlier and thrown it away.

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Trace poly1 by hand ($n = 4, x = 2$)

- Ignore p and the coefficients. Watch one variable: xp

outer i	inner loop runs	xp goes	mults	repeated?
0	never	1	0	—
1	j = 1	1 $\Rightarrow$ 2	1	1 new
2	j = 1, 2	1 $\Rightarrow$ 2 $\Rightarrow$ 4	2	1 new, 1 redone
3	j = 1, 2, 3	1 $\Rightarrow$ 2 $\Rightarrow$ 4 $\Rightarrow$ 8	3	1 new, 2 redone
4	j = 1, 2, 3, 4	1 $\Rightarrow$ 2 $\Rightarrow$ 4 $\Rightarrow$ 8 $\Rightarrow$ 16	4	1 new, 3 redone

Total: 10 multiplications to produce 4 new values. Six of the ten recompute a number the program held one iteration earlier and threw away.

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Solution 2 (carry the power forward)

```
double poly2(const double a[], int n, double x) {
    double p = a[0];
    double xp = 1.0;
    for (int i = 1; i <= n; i++) {
        xp *= x;
        p += a[i] * xp;
    }
    return p;
}
```

Insight: $x^i = x \cdot x^{i-1}$, so carry the power forward instead of rebuilding it

Same output to every digit. Same loop structure at a glance. **A fundamentally different amount of work.**

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Q2-3

- At $n = 1,000$, how many times more multiplications does `poly1` do?

✓ `poly1` performs $\frac{n(n+1)}{2}$ multiplications

✓ `poly2` performs n

- A. About 2x
- B. About 500x
- C. About 1000x
- D. About 1,000,000x
- E. None of the above

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Answer

- B) About 500x

$$\frac{n(n+1)/2}{n} = \frac{n+1}{2} = \frac{1001}{2} \approx 500$$

- ✓ the gap grows with n , it gets worse as the input gets bigger

n	poly1/poly2
100	50x
1,000	500x
1,000,000	500,000x

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Timing

```
#include <iostream>
#include <iomanip>
#include <chrono>
#include <cstdlib>
#include <vector>

int main(int argc, char *argv[]) {
    if (argc != 3) {
        std::cerr << "Usage: " << argv[0] << " <n> <alg>\n";
        return 1;
    }

    int n = std::stoi(argv[1]);
    int alg = std::stoi(argv[2]);
    const double x = 0.999999;

    std::vector<double> a(n + 1); // random coefficients in [-1, 1]
    std::srand(42); // fixed seed = reproducible
    for (int i = 0; i <= n; i++)
        a[i] = 2.0 * ((double) std::rand() / RAND_MAX) - 1.0;

    double p;
    auto start = std::chrono::high_resolution_clock::now();
    if (alg == 1) p = poly1(a.data(), n, x);
    if (alg == 2) p = poly2(a.data(), n, x);
    auto end = std::chrono::high_resolution_clock::now();

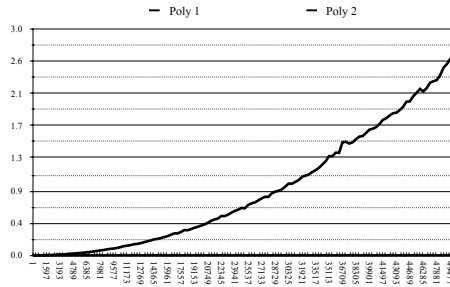
    std::chrono::duration<double> elapsed = end - start;
    std::cout << std::fixed << std::setprecision(10);
    std::cout << p << " " << (double) elapsed.count() << std::endl;

    return 0;
}
```



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Hypothesis, Prediction, Validation



The scientific method, applied to code:

- **Observe:** run the programs across a range of input sizes
- **Hypothesize:** propose a model, e.g. $T(n) = cn^2$
- **Predict:** use the model to forecast a running time you have not yet measured
- **Validate:** measure it. If the prediction holds, the model survives. If not, revise.

A hypothesis earns its keep by predicting a measurement you haven't taken yet.

```
$ seq 1 399 50000 | while read n; do printf "%s\t" "$n"; ./prog $n 1; done > p1.txt
$ seq 1 399 50000 | while read n; do printf "%s\t" "$n"; ./prog $n 2; done > p2.txt
```

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Q2-4

- Consider the same timing program, change one flag and run it again (-O0 => -O3)
 - ✓ not one line of your code changes, same algorithm, same coefficients, the only difference is what the compiler generates
- Fill this table from your own machine:

n	poly1 -O0	poly1 -O3	speedup T_O0 / T_O3
131,072			
262,411			
524,288			

How much faster is -O3?

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Q2-5

- A student asks an AI assistant to “benchmark these two functions.” It produces this code and reports that `poly2` is about 8x faster? What is the most serious flaw in this benchmark?

```
int main() {
    double a[101];
    for (int i = 0; i <= 100; i++) a[i] = 1.0;

    auto t1 = std::chrono::high_resolution_clock::now();
    poly1(a, 100, 0.5);
    auto t2 = std::chrono::high_resolution_clock::now();
    poly2(a, 100, 0.5);
    auto t3 = std::chrono::high_resolution_clock::now();

    std::cout << "poly1: " << (t2-t1).count() << "\n";
    std::cout << "poly2: " << (t3-t2).count() << "\n";
}
```

- It measures at a single input size, so it cannot reveal how either function grows
- The return values are unused, so -O2 may delete both calls entirely
- A single untimed run at small n is dominated by noise and cold-cache effects
- All of the above are genuine flaws
- None of the above, the benchmark is sound

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Answer

- D) All of the above are genuine flaws
 - ✓ A) a measurement at one input size tells you nothing about growth
 - ✓ B) real, and invisible. The return values are discarded. With optimization on, the compiler may notice both calls have no observable effect and remove them, leaving you timing an empty program. The standard fix is to consume the result.
 - ✓ C) real. A single run at small n measures cold caches, page faults, and scheduler noise as much as it measures your code. Repeat and take a median.

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Limitations of empirical analysis

- Implementation challenges
 - ✓ you must implement every algorithm you want to compare, and implementation quality varies
- Experimentation challenges
 - ✓ designing representative test cases; long execution cycles; sensitivity to input distribution
- Environmental dependencies
 - ✓ compiler flags (-O0, -O1, -O2, -O3), HW differences (AVX, branch prediction, cache sizes), OS scheduling, background load
- Reproducibility
 - ✓ timings fluctuate; exact conditions are hard to replicate; results may not generalize beyond the range tested

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